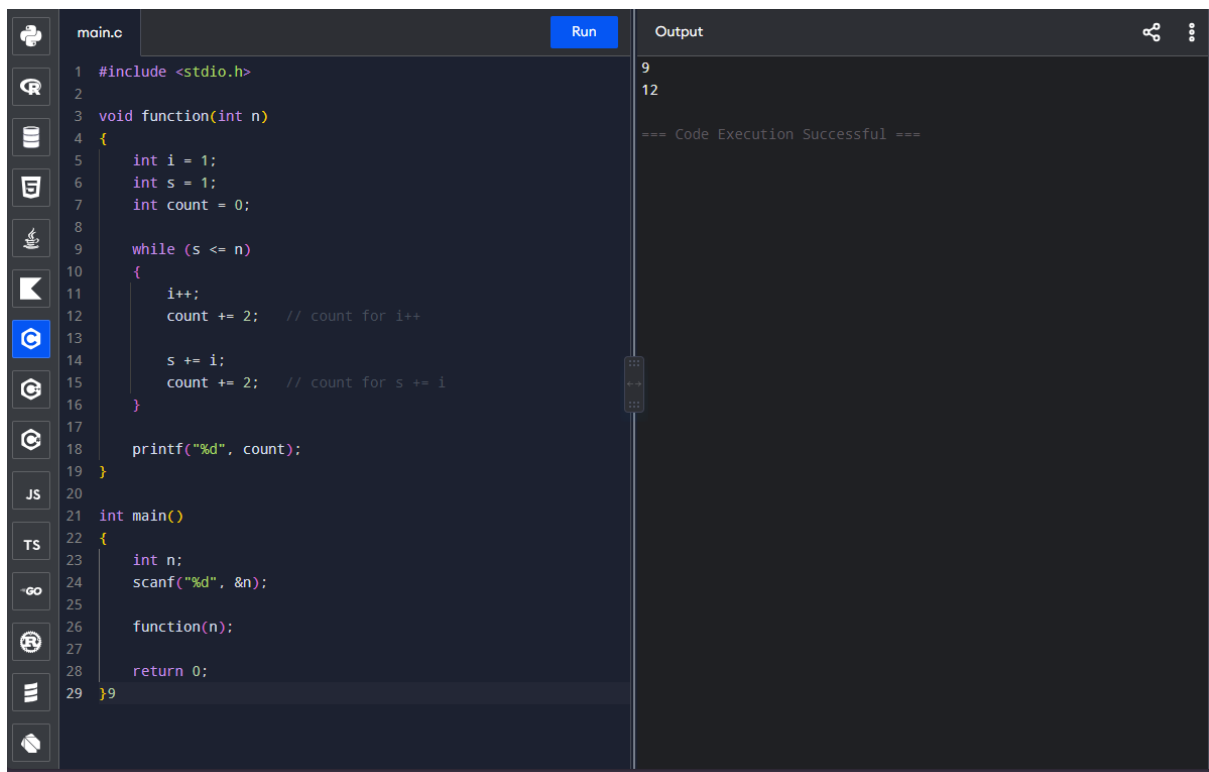


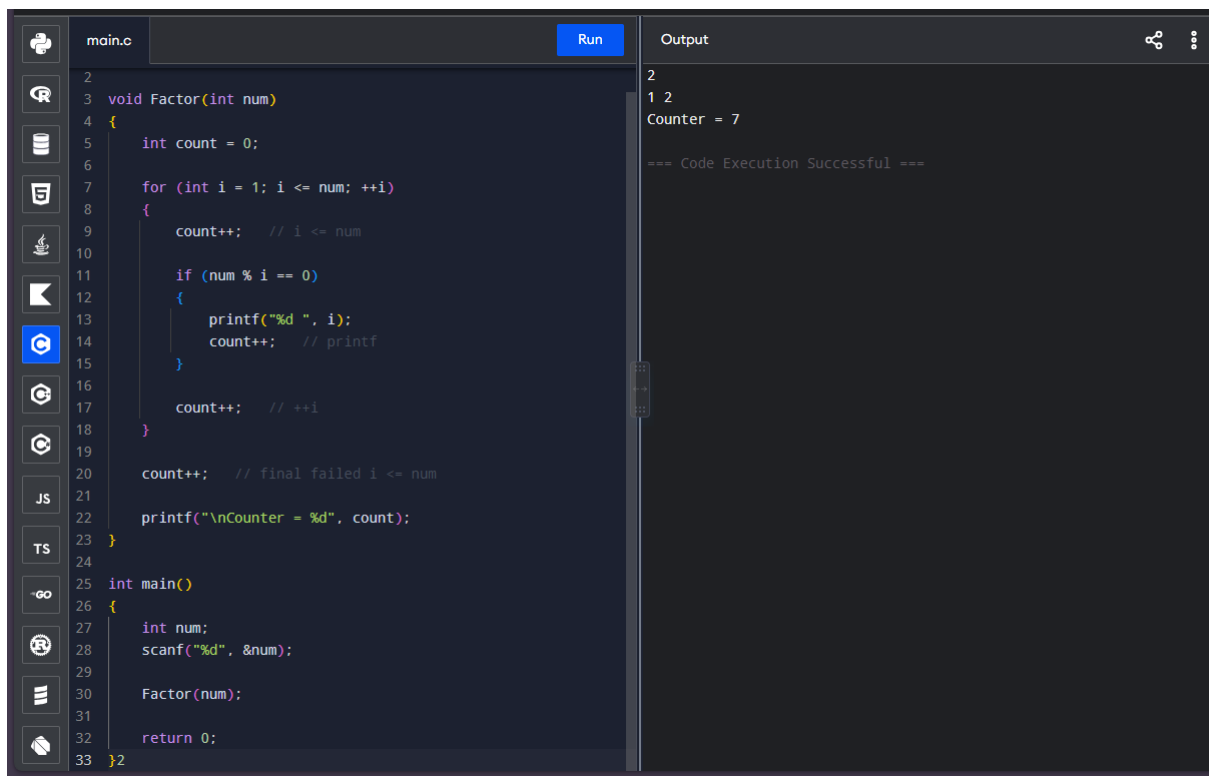
EX-1



```
main.c Run Output
1 #include <stdio.h>
2
3 void function(int n)
4 {
5     int i = 1;
6     int s = 1;
7     int count = 0;
8
9     while (s <= n)
10    {
11        i++;
12        count += 2; // count for i++
13
14        s += i;
15        count += 2; // count for s += i
16    }
17
18    printf("%d", count);
19 }
20
21 int main()
22 {
23     int n;
24     scanf("%d", &n);
25
26     function(n);
27
28     return 0;
29 }
```

9
12
=== Code Execution Successful ===

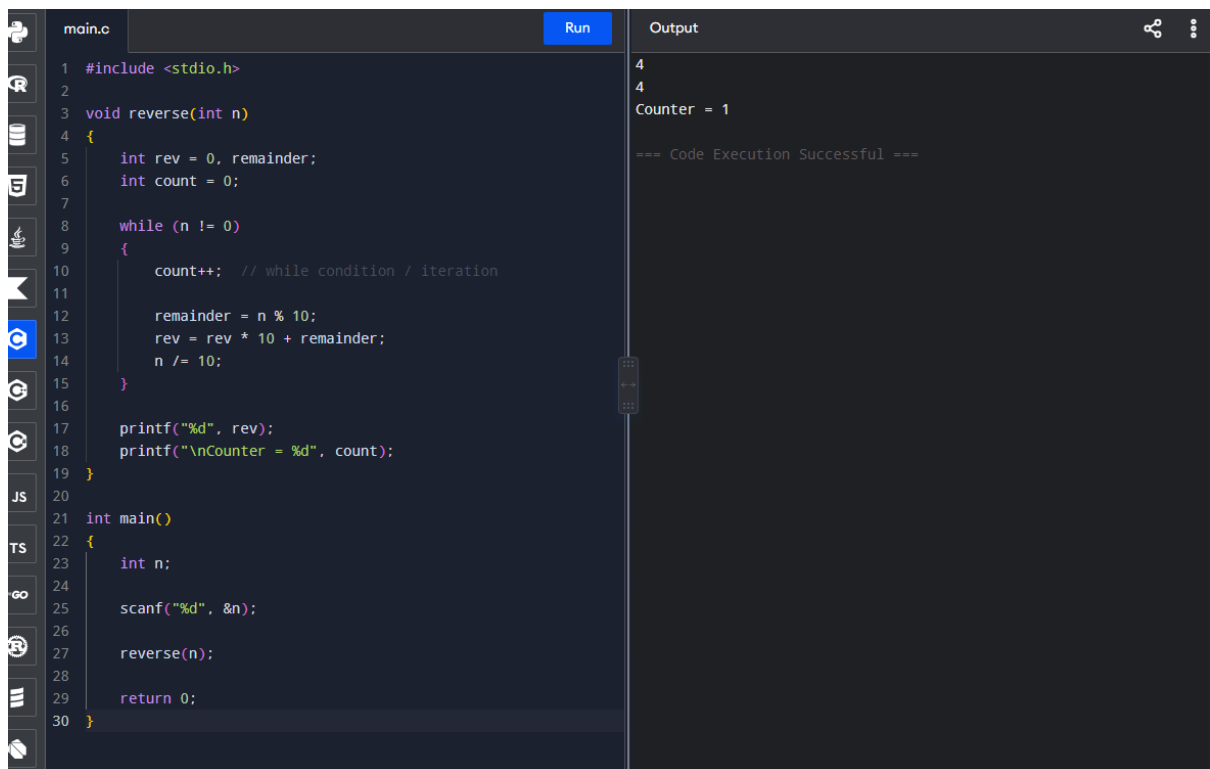
EX-3



```
main.c Run Output
2
3 void Factor(int num)
4 {
5     int count = 0;
6
7     for (int i = 1; i <= num; ++i)
8     {
9         count++; // i <= num
10
11         if (num % i == 0)
12         {
13             printf("%d ", i);
14             count++; // printf
15         }
16
17         count++; // ++i
18     }
19
20     count++; // final failed i <= num
21
22     printf("\nCounter = %d", count);
23 }
24
25 int main()
26 {
27     int num;
28     scanf("%d", &num);
29
30     Factor(num);
31
32     return 0;
33 }
```

2
1 2
Counter = 7
=== Code Execution Successful ===

EX-2



The screenshot shows a C code editor with a file named `main.c`. The code implements a function `reverse` that takes an integer `n` and returns its reverse. The function uses a `while` loop to extract digits from `n` and build the reversed number `rev`. The `main` function prompts the user to enter a number, calls `reverse`, and prints the result along with a counter.

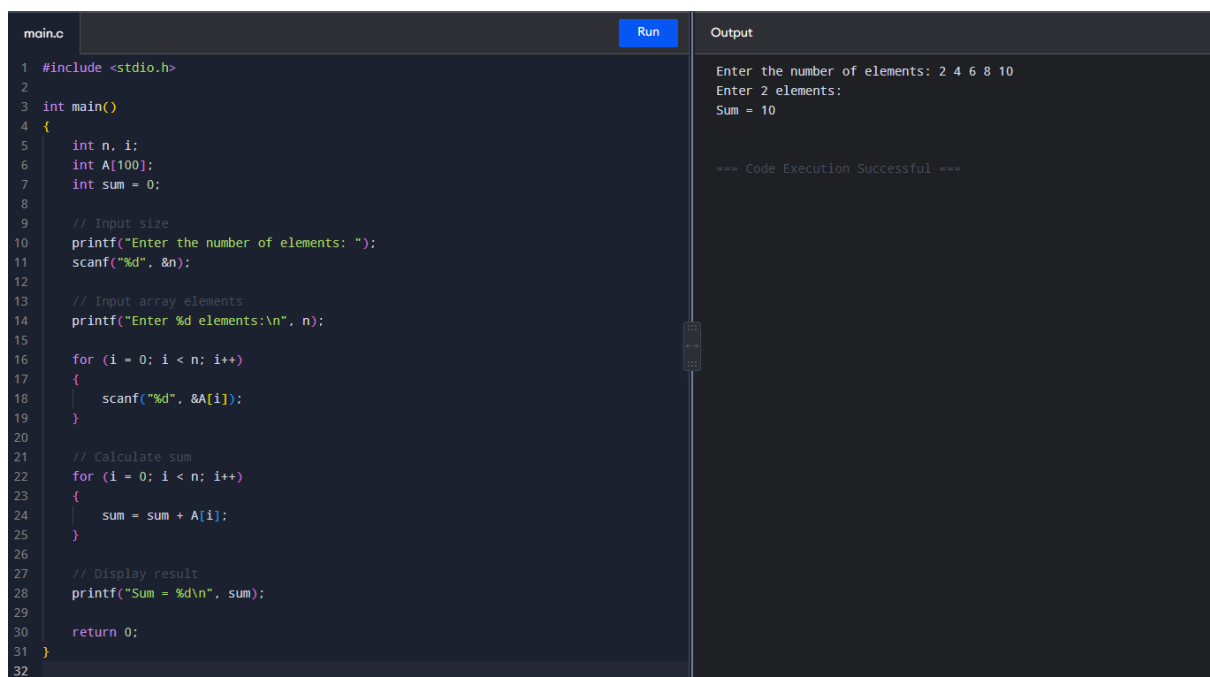
```
1 #include <stdio.h>
2
3 void reverse(int n)
4 {
5     int rev = 0, remainder;
6     int count = 0;
7
8     while (n != 0)
9     {
10         count++; // while condition / iteration
11
12         remainder = n % 10;
13         rev = rev * 10 + remainder;
14         n /= 10;
15     }
16
17     printf("%d", rev);
18     printf("\nCounter = %d", count);
19 }
20
21 int main()
22 {
23     int n;
24
25     scanf("%d", &n);
26
27     reverse(n);
28
29     return 0;
30 }
```

The output window shows the following text:

```
4
4
Counter = 1

=== Code Execution Successful ===
```

EX-4



The screenshot shows a C code editor with a file named `main.c`. The code implements a program that calculates the sum of elements in an array. It prompts the user to enter the number of elements and the elements themselves, then calculates and displays the sum.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i;
6     int A[100];
7     int sum = 0;
8
9     // Input size
10    printf("Enter the number of elements: ");
11    scanf("%d", &n);
12
13    // Input array elements
14    printf("Enter %d elements:\n", n);
15
16    for (i = 0; i < n; i++)
17    {
18        scanf("%d", &A[i]);
19    }
20
21    // Calculate sum
22    for (i = 0; i < n; i++)
23    {
24        sum = sum + A[i];
25    }
26
27    // Display result
28    printf("Sum = %d\n", sum);
29
30    return 0;
31 }
32
```

The output window shows the following text:

```
Enter the number of elements: 2 4 6 8 10
Enter 2 elements:
Sum = 10

=== Code Execution Successful ===
```

EX-5

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3 int main() 4 { 5 int n, i; 6 int count = 0; 7 8 printf("Enter n: "); 9 scanf("%d", &n); 10 11 for (i = 0; i < n; i++) 12 { 13 count++; 14 } 15 16 printf("Count = %d\n", count); 17 18 return 0; 19 } 20</pre>		<pre>Enter n: 5 Count = 5 --- Code Execution Successful ---</pre>